

Zachary Benson

Senior full stack engineer (ASP.NET, .NET, C#, React, Angular, Node, AWS, TypeScript)

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PROFESSIONAL SUMMARY

Versatile and results-oriented **Senior Full-Stack Engineer** with **8+ years of experience** architecting and delivering high-performance, enterprise-grade applications within the **Microsoft ecosystem**. Expert in the **.NET stack (C#, .NET 8/6, ASP.NET Core)** and modern frontend ecosystems including **React, Angular, and Blazor**.

Deeply skilled in building robust **Microservices architectures**, secure **Web APIs**, and modular desktop/mobile solutions using **WPF and .NET MAUI**. Proven track record in optimizing full-stack performance through **Entity Framework Core** profiling, asynchronous programming, and advanced **SSR/SSG strategies**.

Extensive experience managing cloud-native infrastructure on **Microsoft Azure (AKS, App Services, Service Bus)**, leveraging **Docker** and **Kubernetes** for containerization, and implementing high-velocity **CI/CD pipelines via Azure DevOps and GitHub Actions**. A strong advocate for **Clean Architecture, SOLID principles**, and **TDD**, with a background in leading Agile/Scrum teams, mentoring developers, and aligning sophisticated technical roadmaps with core business objectives.

EDUCATION

University of Florida - Bachelor's Degree in computer Engineering, 2014 - June 2018

TECHNICAL SKILLS

Frontend & Desktop Frameworks: Advanced expertise in React.js, Next.js, and Angular for web; and WPF (Windows Presentation Foundation), WinUI 3, and UWP for high-performance desktop applications. Proficient in building SPAs and native Windows apps with a focus on accessibility, performance, and responsive UI.

Mobile Development: Specialist in cross-platform mobile architecture using .NET MAUI and Xamarin.Forms/Native. Expert in building shared-codebase applications for iOS and Android while utilizing Native Embeddings and Custom Renderers/Handlers for platform-specific hardware integration.

State Management & Architecture: Proficient in Redux (Toolkit) and Context API for web, alongside the MVVM (Model-View-ViewModel) pattern for desktop and mobile development. Expert in utilizing CommunityToolkit, Mvvm, ReactiveUI, and Prism for scalable, maintainable architectures.

Component Libraries & Design Systems: Skilled in customizing Fluent UI (Web and WinUI), Syncfusion, and Telerik. Experienced in building shared UI component libraries and design systems for enterprise-scale reuse across Web and Desktop environments.

Backend & Microservices Development: Expertise in C#, .NET 8/6, ASP.NET Core Web API, and Entity Framework (EF) Core for scalable backend architecture. Deep experience with Event-Driven Microservices, Clean Architecture, and Domain-Driven Design (DDD) patterns.

API Architecture: Architected and consumed RESTful, GraphQL, and gRPC APIs; integrated AutoMapper, Swagger/OpenAPI, and SignalR for real-time bi-directional communication between backend services and desktop/mobile clients.

Cloud & DevOps (Azure Focus): Deep hands-on experience with Azure Services (Functions, App Service, Cosmos DB, and Service Bus). Proficient in Terraform and Bicep for IaC. Built robust CI/CD pipelines using Azure DevOps and GitHub Actions, including automated builds and distribution for App Center and Windows Store.

Messaging & Event Streaming: Experience with Azure Service Bus, Azure Event Hubs, and RabbitMQ to enable asynchronous communication and real-time data streaming in distributed enterprise systems.

Testing & Quality Assurance: Advanced testing with xUnit, Moq, and FluentAssertions for backend; Jest and Playwright for web; and Appium or XCUITest/Espresso for mobile automated testing.

Database Systems: Hands-on with Microsoft SQL Server (T-SQL), Azure SQL Database, and SQLite (for local mobile/desktop persistence). Proficient with EF Core and Dapper for high-performance data schemas and migrations.

Security & Auth: Implemented Microsoft Identity Platform (MSAL), Azure AD / Entra ID, OAuth2, and JWT. Enforced RBAC, CORS, and secure API gateways using Azure API Management (APIM).

Agile Practices & Collaboration: Strong experience in Agile/Scrum and SAFe environments. Collaborated via Azure Boards and Microsoft Teams across globally distributed teams. Led architecture reviews following SOLID and DRY principles.

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WORK EXPERIENCE

Deloitte (Remote)- C# Full Stack Engineer

Jul 2024 – Present

- Architected and delivered a high-performance, cloud-native e-commerce platform using the **.NET stack (ASP.NET Core, SQL Server, React)** within a microservices architecture, ensuring scalability, modularity, and independent deployments.
- Led the design and architecture of **RESTful and GraphQL APIs** using **C# and Hot Chocolate**, establishing consistent patterns for versioning, error handling, authentication (**IdentityServer, JWT, OAuth2**), and data pagination.
- Created a centralized **API Gateway pattern** with rate limiting, request validation, and caching using **Redis and Azure API Management** to improve latency and throughput across distributed services.
- Designed and built a highly scalable **React component architecture using TypeScript**, following atomic design principles and Storybook-driven development, supporting reusable, testable, and responsive UI components across multiple brands and devices.
- Implemented **Next.js SSR/SSG strategies**, dynamic imports, lazy loading, and code splitting to enhance frontend performance, Core Web Vitals, and SEO readiness.
- Defined and maintained backend modules with **ASP.NET Core**, applying **Domain-Driven Design (DDD)**, dependency injection, and **Entity Framework (EF) Core**, improving maintainability and separation of concerns.
- Integrated secure payment workflows with **Stripe and PayPal**, ensuring PCI-DSS compliance, and built extensible **C# middleware** for webhook handling and fraud detection.
- Built and deployed cloud infrastructure using **Azure (App Service, Blob Storage, Azure SQL, Functions)** and **Terraform/Bicep** for Infrastructure as Code (IaC), supporting scalability, monitoring, and disaster recovery.
- Automated full CI/CD pipelines using **Azure DevOps (Pipelines)**, Docker, Helm, and **Azure Kubernetes Service (AKS)**, enabling faster deployments with environment parity and canary releases.
- Implemented full-stack observability using **Application Insights, ELK Stack, and Azure Monitor**, including dashboards, alerts, and logging standards for production insights.
- Applied **test-driven development (TDD)** using **xUnit, Moq, Cypress**, and React Testing Library, ensuring robust test coverage for unit, integration, and E2E flows.
- Collaborated with designers to translate **Figma** wireframes into responsive, accessible, and scalable UI systems, promoting consistent branding and usability across platforms.
- Authored technical documentation, **API specifications (OpenAPI/Swagger)**, component libraries, and DevOps runbooks to support ongoing development and team knowledge sharing.
- Mentored frontend and backend engineers on **Clean Architecture**, code quality, system performance, and agile delivery practices, contributing to a high-performing, cross-functional team.

General Motors (Florida) – Senior Full Stack Engineer

Oct 2022 - May 2024

- Architected and developed a **real-time vehicle tracking platform** using the **Microsoft stack (ASP.NET Core, SQL Server, Angular)**, supporting thousands of **concurrent users** and delivering accurate **live telemetry** and **fleet performance insights**.
- Designed and deployed **cloud-native infrastructure on Azure**, leveraging **Blob Storage, Azure Functions, API Management, App Service, and Entra ID (Azure AD)**, ensuring **secure authentication** and **highly available backend services**.
- Engineered **real-time data streaming systems** using **SignalR and WebSockets**, enabling **instant map updates, route tracking, and telemetry broadcasting** with **low-latency data handling** in a **distributed .NET environment**.
- Developed **secure authentication and authorization layers** using **Microsoft Identity Platform, OAuth2, JWT, and Multi-Factor Authentication (MFA)**, enhancing **system resilience** and **data protection**.
- Built **scalable, reusable UI components and Directives** with **Angular + TypeScript**, implementing **advanced state management patterns** with **NgRx and Services** to ensure **consistent and performant frontend behavior**.
- Designed and implemented **RESTful APIs and GraphQL endpoints (Hot Chocolate)** for **modular data consumption**, aligning backend services with **scalable microservice communication models** using **C#**.
- Optimized **SQL Server performance** using **advanced indexing, partitioning, execution plan analysis, and Entity Framework (EF) Core**, improving **data retrieval speeds** and reducing **query load** under high traffic.
- Implemented **CI/CD pipelines** using **Azure DevOps (Pipelines), GitLab CI, and Docker**, ensuring **automated testing, build, and zero-downtime deployment** across staging and production environments.
- Enabled **infrastructure-as-code (IaC)** practices using **Terraform and Bicep**, standardizing **Azure environment provisioning** and improving **deployment consistency**.
- Led development of **interactive dashboards** using **custom charting libraries and Map APIs**, utilizing **Angular's change detection strategies** and **RxJS Observables** for **high-frequency visual analytics**.
- Applied **performance tuning techniques** such as **Angular Route Guard lazy loading, OnPush change detection, and distributed caching (Redis)**, significantly reducing **initial load times**.
- Authored and maintained **comprehensive technical documentation, API references, architectural diagrams, and operational guides** for development and DevOps teams.
- Championed **Agile and Scrum methodologies**, conducting **sprint planning and code reviews** while **mentoring junior engineers** in **SOLID principles, Clean Architecture, and C# best practices**.
- Conducted extensive **automated and manual testing** using **xUnit, Moq, Jasmine, and Karma**, ensuring **robust QA coverage** across backend and frontend services.

Zillow (Remote) – C# Full Stack Engineer

Mar 2020 - Sep 2022

- Designed and developed **enterprise-level real estate platforms** using the **Microsoft stack (ASP.NET Core, SQL Server, React)**, delivering intuitive **property search, listings, and management systems**.
- Architected and deployed **cloud-native infrastructure on Azure**, leveraging **Blob Storage, Azure Functions, and API Management**, ensuring **fault-tolerant, scalable backend services** and **efficient serverless data workflows**.
- Developed **advanced frontend features** such as **interactive map visualizations, dynamic filters, and real-time property listings**, enhancing **UX** across web and mobile interfaces using **React and TypeScript**.
- Built and secured **RESTful APIs** for **real-time data access** and **third-party integrations**, including **payment gateways, geolocation services, and MLS property data providers** using **C#**.
- Implemented **SSR (Server-Side Rendering)** with **Next.js and ASP.NET Core**, **lazy loading**, and **intelligent caching strategies** to significantly improve **SEO and application performance**.
- Optimized **complex client-side state** using **Redux and Context API**, enabling **responsive UI behavior** and **consistent user session experiences**.
- Integrated **TypeScript** and enforced **modern development standards**, improving **maintainability, scalability, and type safety** across the **full .NET/React stack**.
- Collaborated with product owners and UI/UX teams under **Agile and Scrum methodologies** to prioritize **backlogs**, estimate **workloads**, and meet **sprint deliverables** within **Azure Boards**.
- Led **code reviews** and provided **technical mentorship** on **Clean Architecture, SOLID principles**, and **efficient component design patterns**.
- Authored **detailed system documentation, architectural flow diagrams, and deployment guides** to support **onboarding, scalability, and cross-team alignment**.
- Automated **testing workflows** using **xUnit, Moq, Cypress, and Postman**, ensuring **high code coverage, regression protection, and reliable production releases**.
- Initiated **performance audits** and implemented **SQL Server database indexing, query optimization via Entity Framework Core, and Azure CDN integration**, reducing **latency** across key workflows.

TePe USA (California) – Full Stack Engineer

Sep 2018 – Feb 2020

- Architected **high-scale enterprise healthcare platforms** using **React JS and TypeScript**, focusing on **reusable UI components** and **modular design patterns**.
- Leveraged **Material UI** for consistent **CSS styling**; engineered complex **Menus and Drawer components** using **TypeScript** for **strict type safety**.

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- Integrated the **html2canvas npm package** to provide advanced **client-side control** over **image rendering** and **screen capture functionality**.
- Implemented **robust identity management** and **secure user authentication flows** utilizing the **Auth0 npm package**.
- Engineered **high-performance backend services** using **ASP.NET Core Web API**, adhering to **DDD (Domain-Driven Design)** and **SOLID** for **clean separation of concerns**.
- Managed **relational data schemas** with **MS SQL Server** and utilized **EF 6.0 (Entity Framework)** for **database control** and **object-relational mapping**.
- Streamlined **data transfer** using the **AutoMapper NuGet package**; integrated **Swashbuckle** to generate **interactive Swagger UI documentation**.
- Conducted rigorous **code quality audits** and **performance profiling** using **jsHint** and **YSlow** to **optimize frontend delivery**.
- Managed the **full deployment lifecycle** for frontend projects hosted on **DigitalOcean**, utilizing **droplets** for **high availability**.
- Applied **TDD (Test-Driven Development) principles** using **Jest** and **React Testing Library** to ensure **high code coverage** for critical business logic.